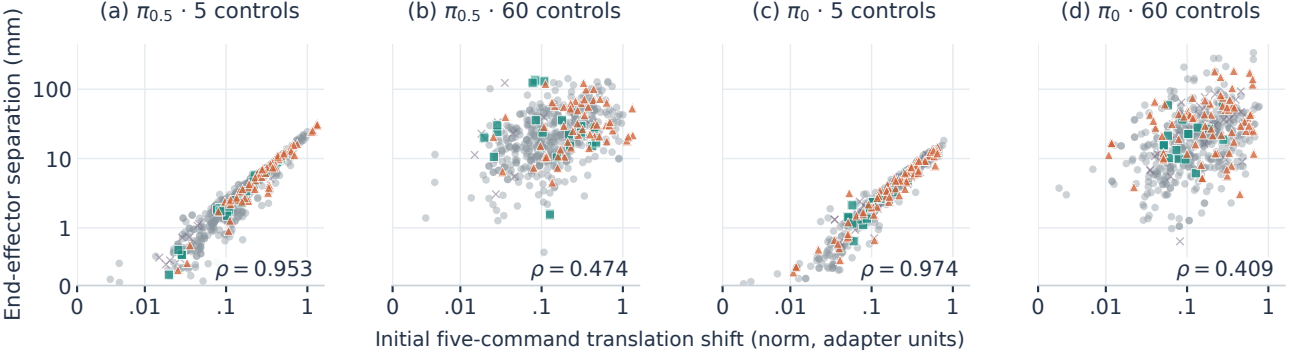

(a) $\pi_{0.5}$ · 5 controls
(b) $\pi_{0.5}$ · 60 controls
(c) π_0 · 5 controls
(d) π_0 · 60 controls
End-effector separation (mm)
100
10
1
0
$\rho = 0.953$
$\rho = 0.474$
$\rho = 0.974$
$\rho = 0.409$
0
.01
.1
1
Initial five-command translation shift (norm, adapter units)
Joint success
Harm
Recovery
Joint failure